

Regulatory Writing & Intelligence

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 01 of 8 · draft and assemble submission content from grounded evidence — a qualified human signs off

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM SEARCHING TO SUBMITTING

Regulatory Writing & Intelligence

A governed medical-writing copilot that retrieves approved evidence, drafts and assembles CTD/eCTD sections with every figure traceable to source — while a qualified regulatory writer reviews and signs off on what is submitted.

\$2.6B

average capitalized cost per approved drug
— the value behind every submission

[gov/peer-reviewed — DiMasi/Tufts]

~\$60M / mo

NPV lost per month of submission delay for a
\$1B-peak asset

[industry-research — McKinsey 2025]

~55%

first-draft CSR hours cut (180→80) with 50%
fewer errors (Merck)

[vendor — Merck-reported]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Medical writers spend disproportionate time retrieving and reconciling evidence rather than reasoning.
- A hallucinated number in a submission is a data-integrity defect — FDA's own gen-AI tool was reported to fabricate studies.
- Only ~13% of top pharma automate table/listing/figure assembly at scale; the rest is manual.
- Every regulated artifact carries 21 CFR Part 11 e-signature and ALCOA+ obligations before the first line of agent code.

THE COST OF DOING NOTHING

~\$60M / month

Industry: ~\$60M NPV lost per month of submission delay for a \$1B-peak asset (~\$180M for a 12-week slip). The FY2026 PDUFA filing fee alone is \$4.68M.

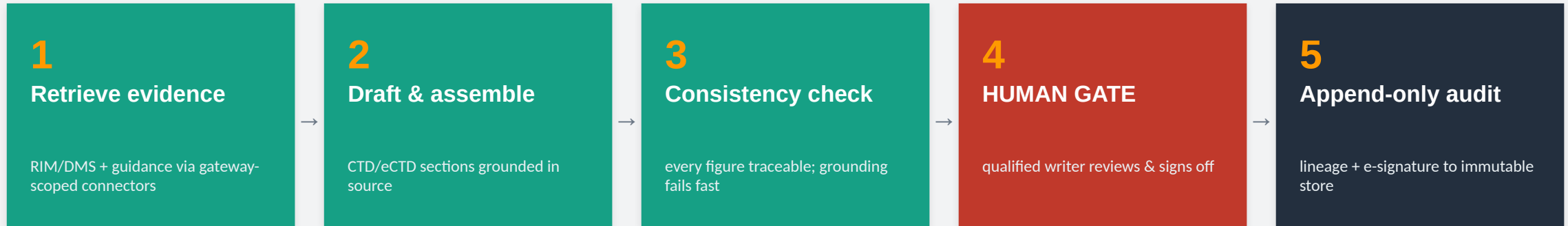
- Data-integrity findings and refuse-to-file risk from inconsistent or unsourced figures.
- Reviewer time lost reconciling drafts assembled by hand across documents.

[industry-research — McKinsey 2025; FDA PDUFA FY2026]

The design line: the agent retrieves, drafts, and checks consistency; a qualified regulatory/medical writer reviews and signs off — nothing is submitted unedited.

How We Solve It — A Governed Pipeline

Retrieve approved evidence through a scoped gateway, draft and assemble grounded sections, verify every figure — then stop at a human gate before anything is filed.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

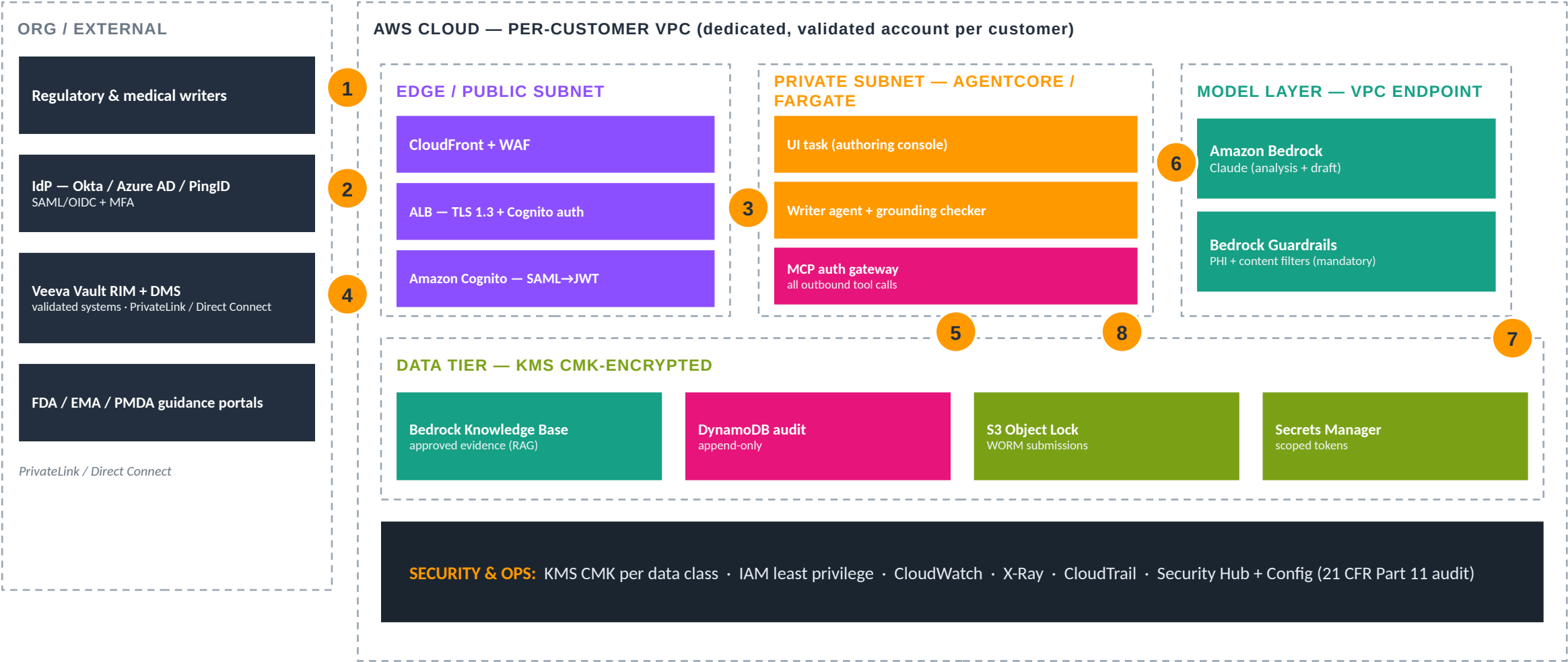
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides what gets submitted

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 writer sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT — no credentials in AWS) 3 authenticated session to authoring console 4 RIM/DMS + guidance read over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails endpoint 7 every draft, source, and e-signature persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~55%

first-draft CSR hours cut, 50% fewer errors
(Merck)

[vendor — Merck]

\$4.68M

FY2026 PDUFA filing fee at stake per
submission

[gov/peer-reviewed — FDA]

~40%

CSR cycle-time reduction in early gen-AI
pilots

[industry-research — McKinsey]

100%

of figures traceable to source by design
(grounding gate)

[design property]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK per data class + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation with role scoping (writer / reviewer / QP).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect Veeva Vault RIM/DMS; index approved guidance into a Knowledge Base.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL sign-off test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway + native/container agent; point connectors at Veeva Vault; index approved guidance.

aws-native-reference/01-regulatory-writing/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.